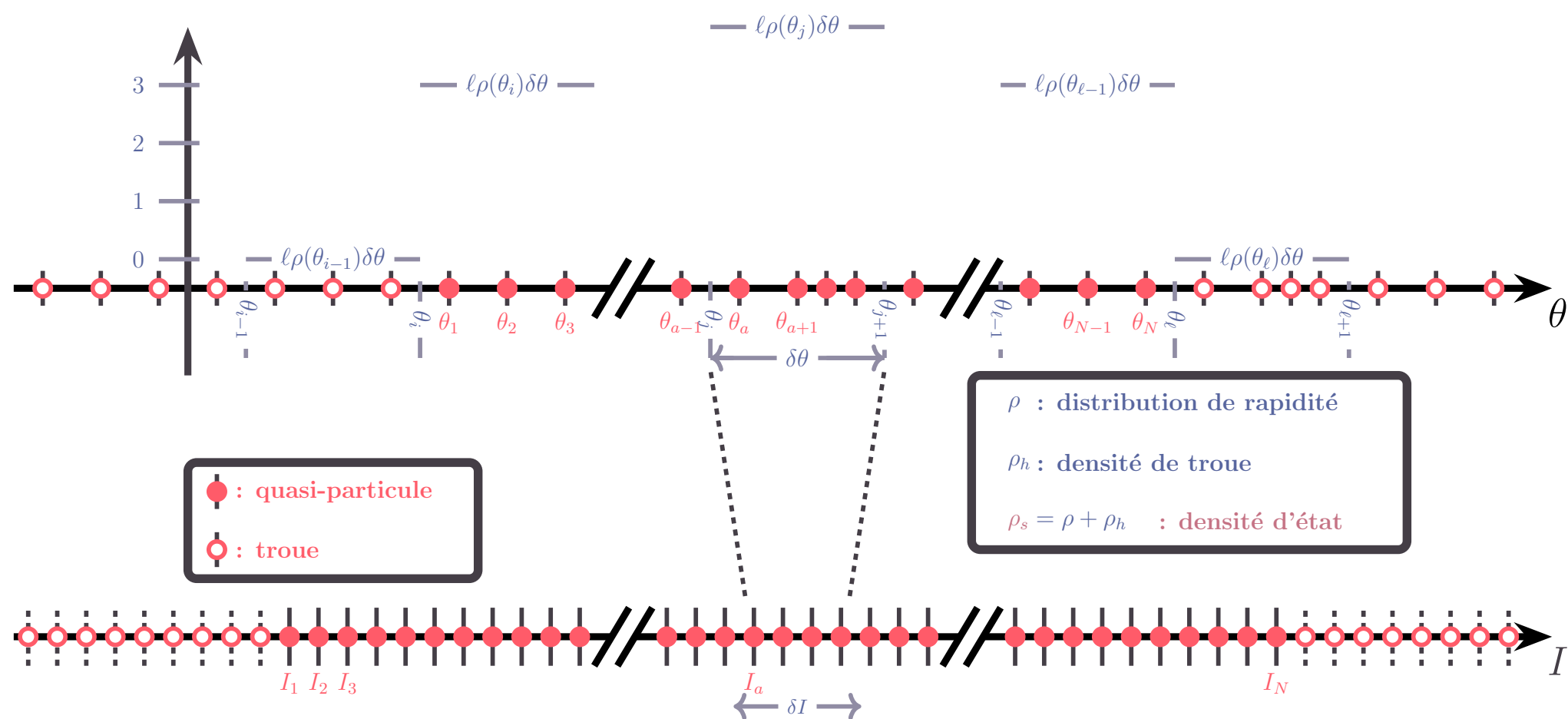
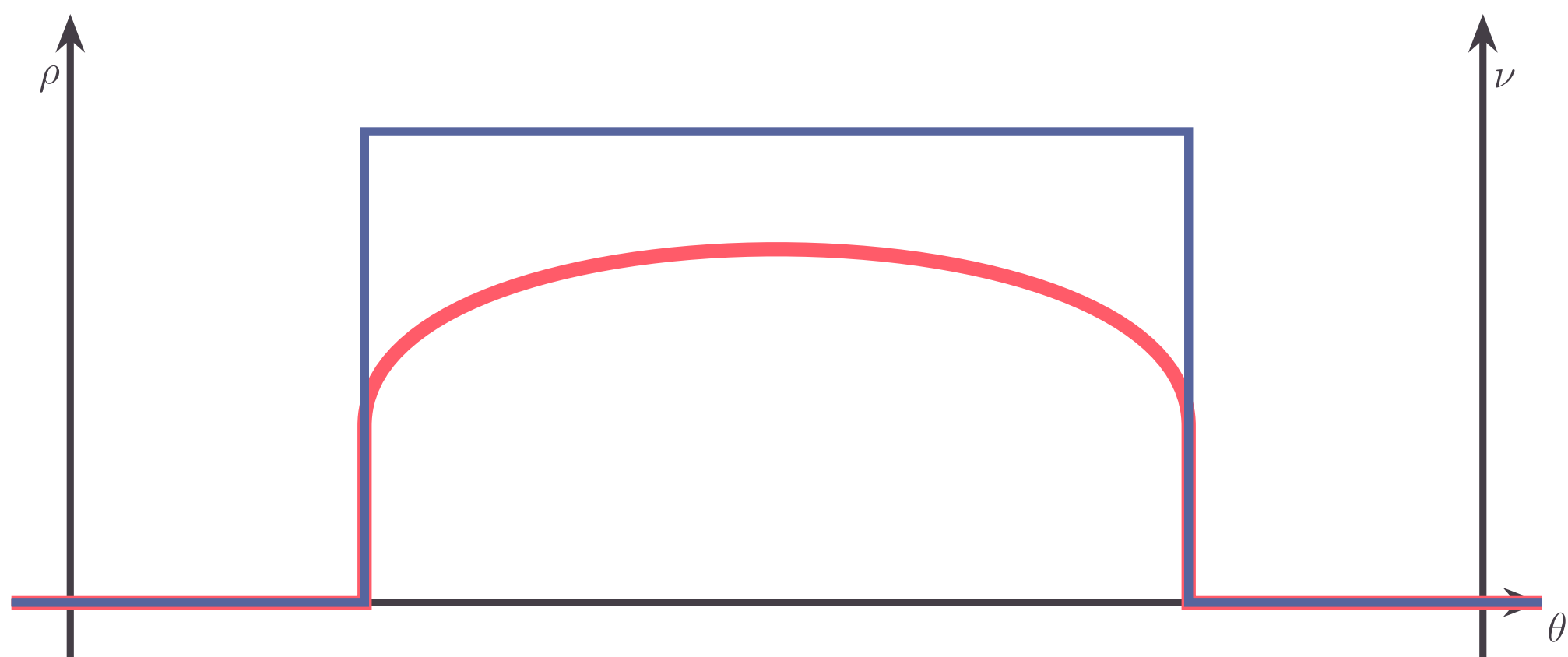
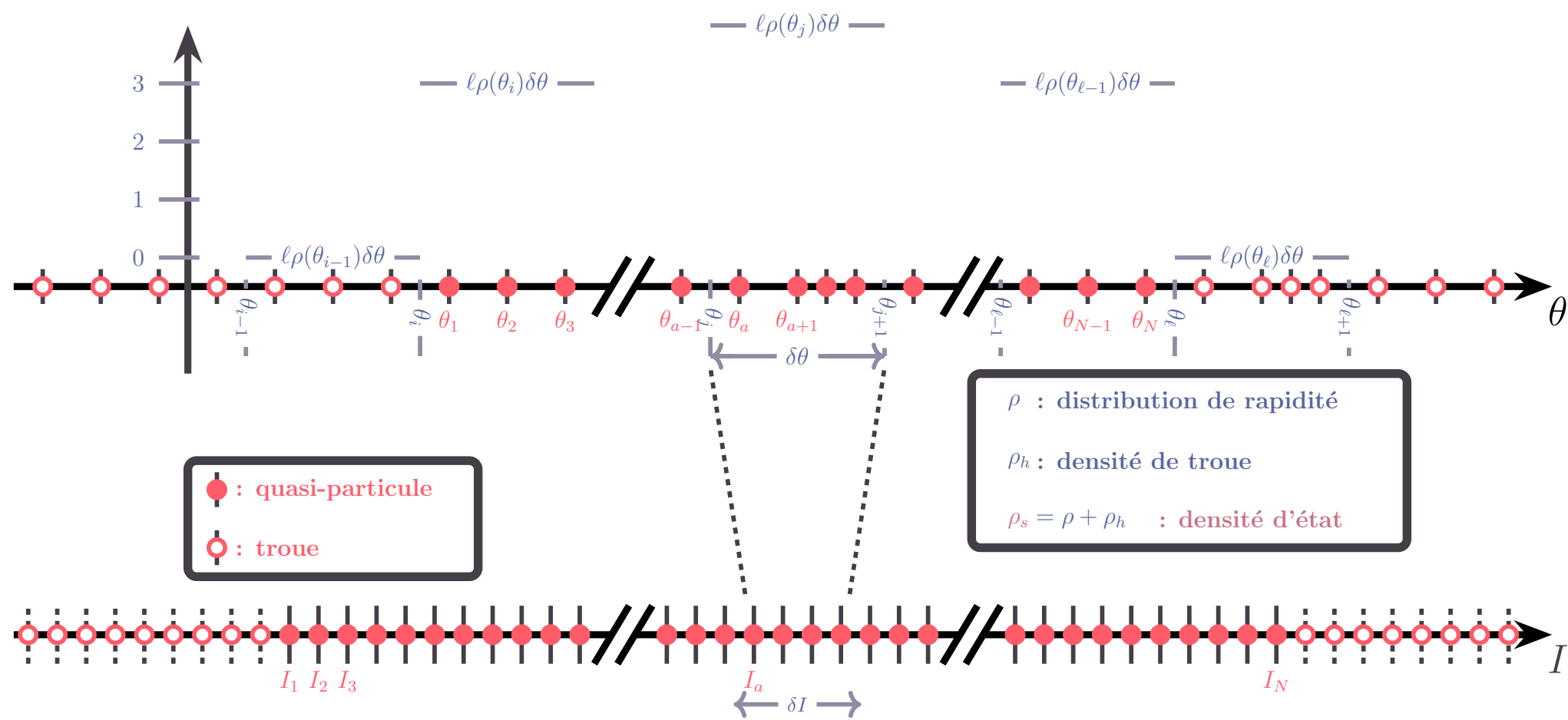
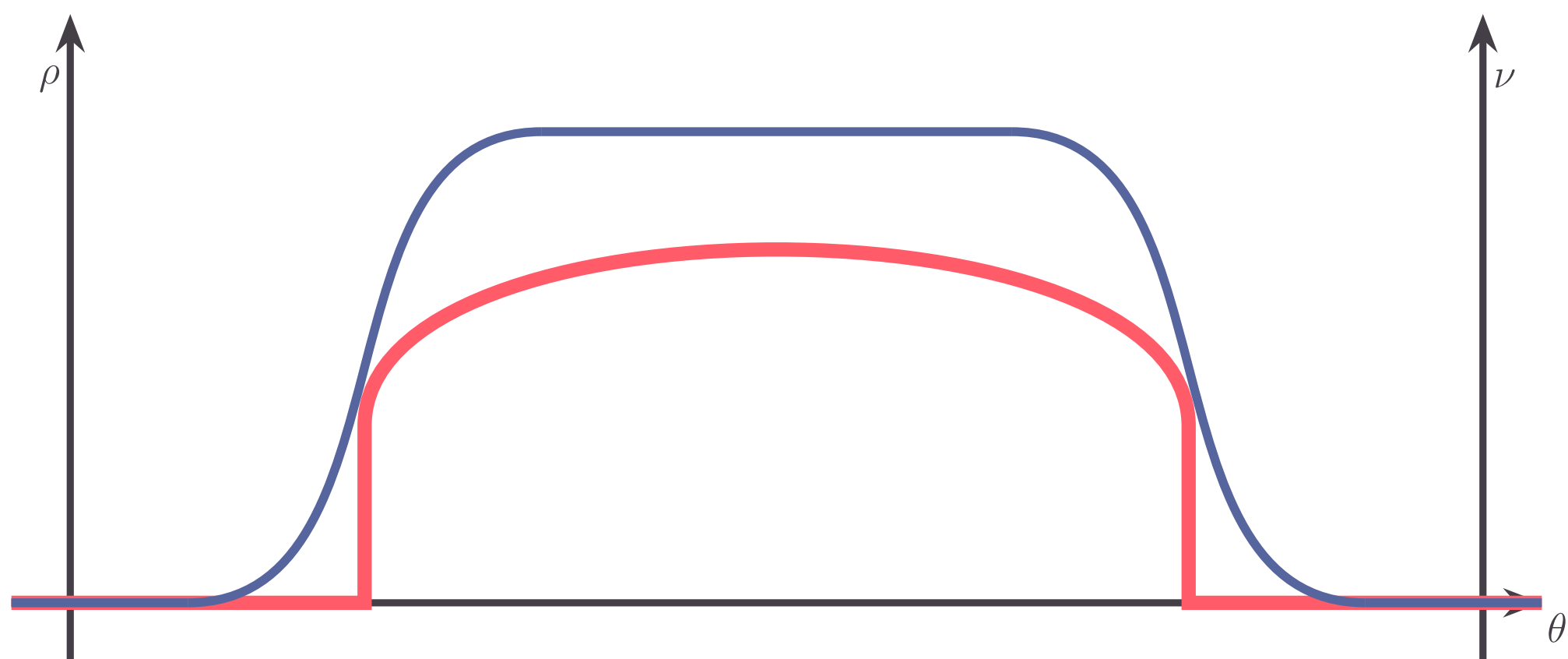
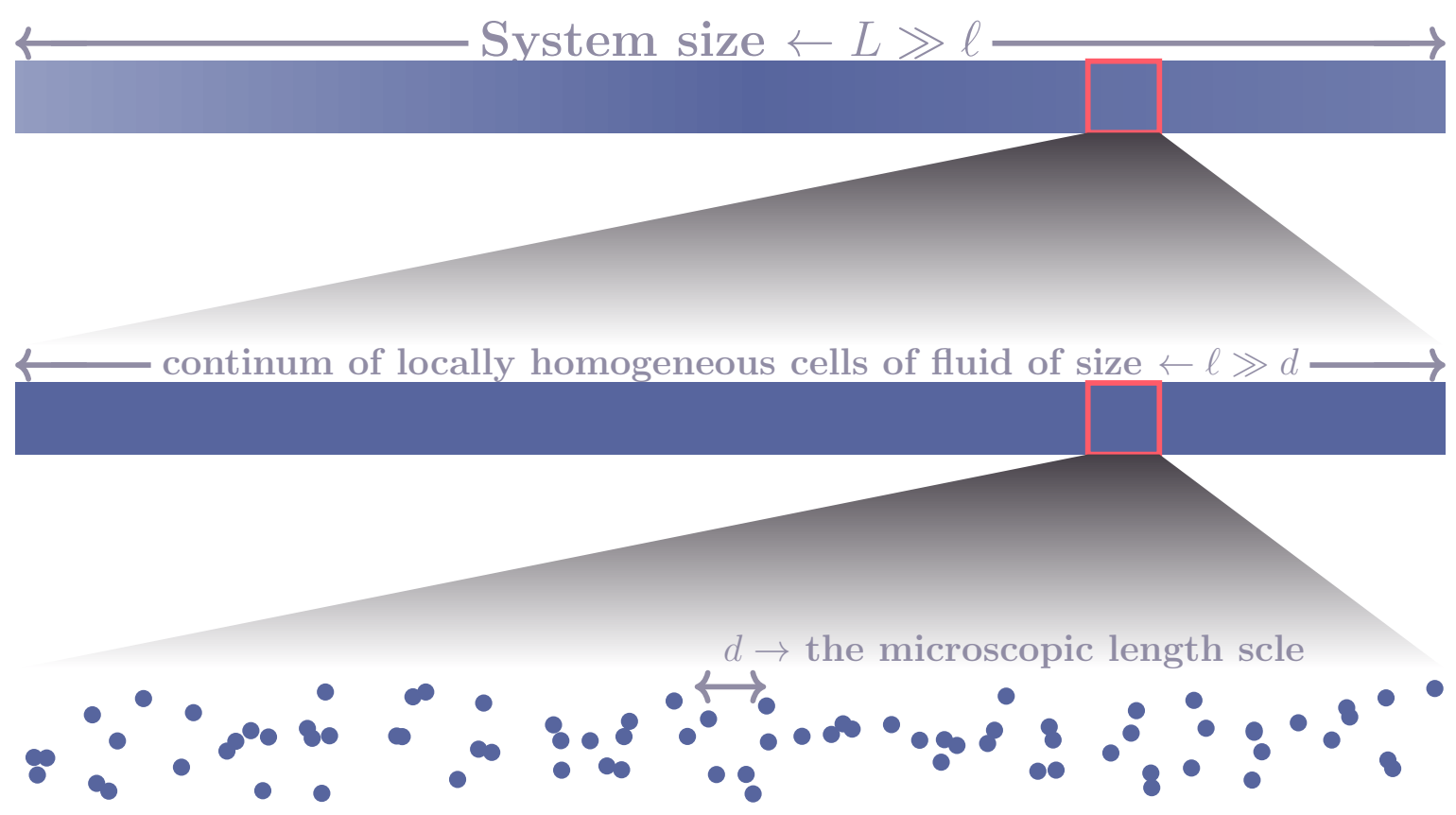
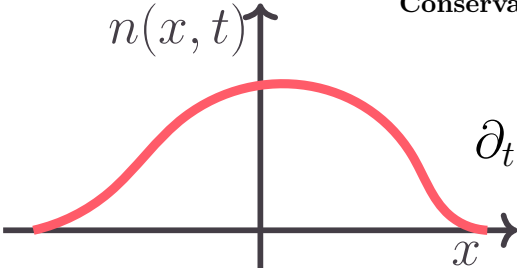






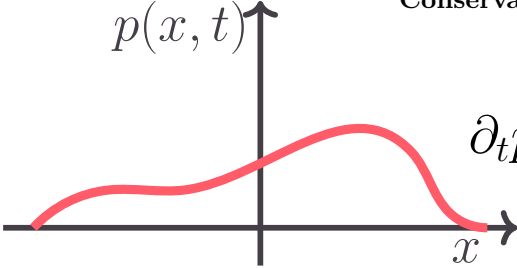



$$n(x, t)$$

Conservation of the atom number :

$$\int dx n(x, t) = N$$

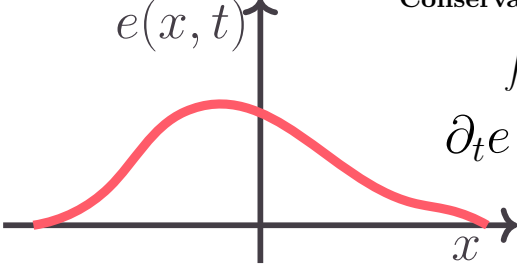
$$\partial_t n + \partial_x j_n = 0$$


$$p(x, t)$$

Conservation of total momentum:

$$\int dx p(x, t) = P$$

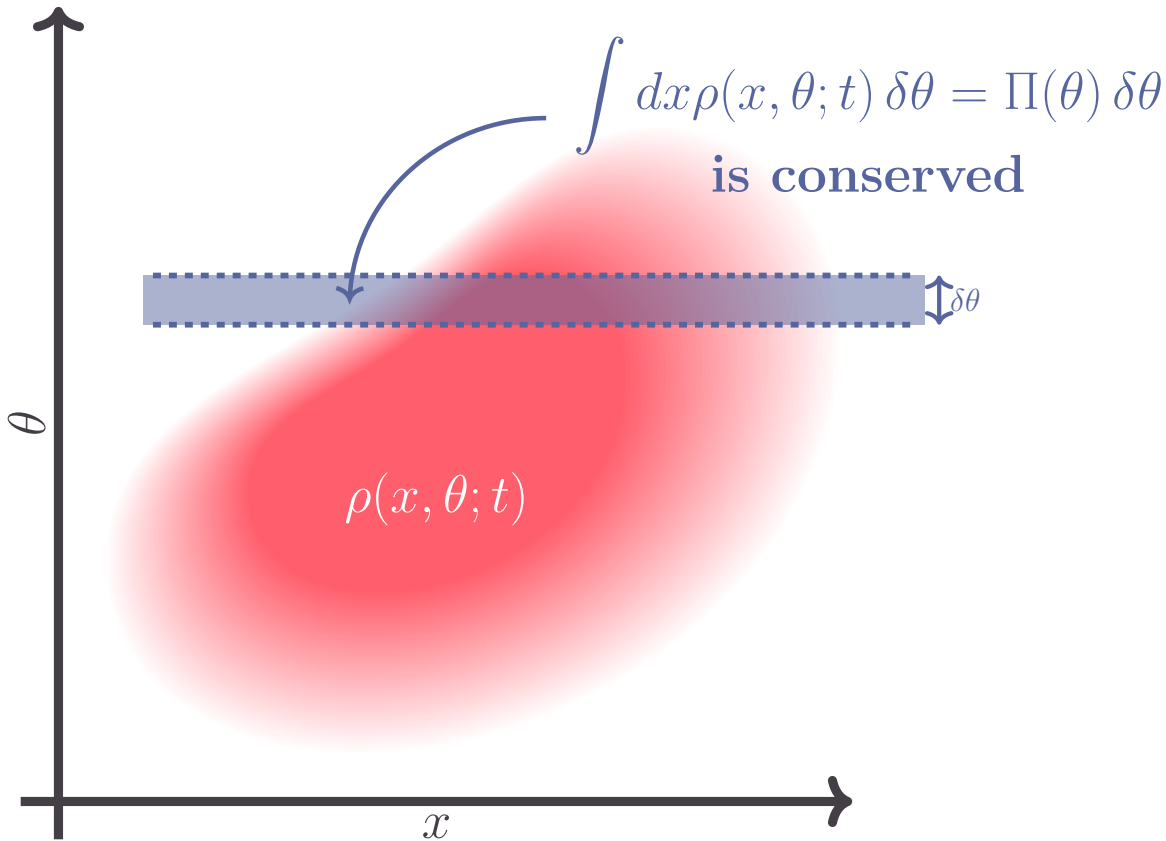
$$\partial_t p + \partial_x j_p = 0$$

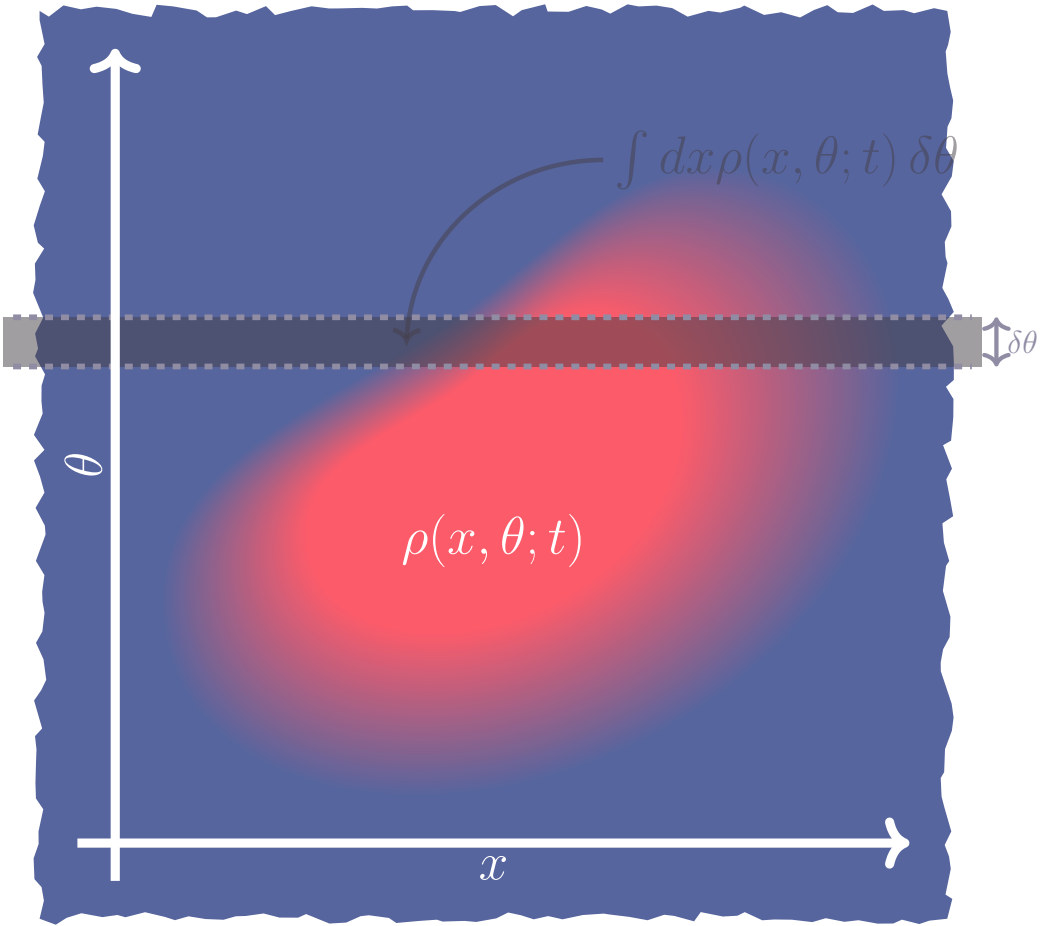

$$e(x, t)$$

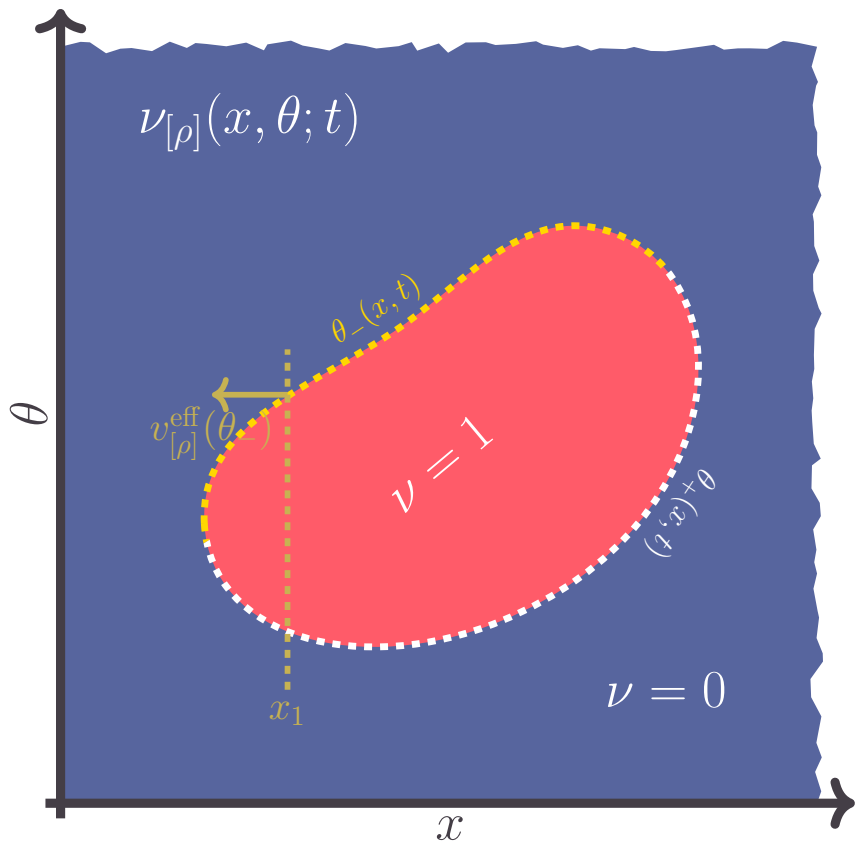
Conservation of total energy :

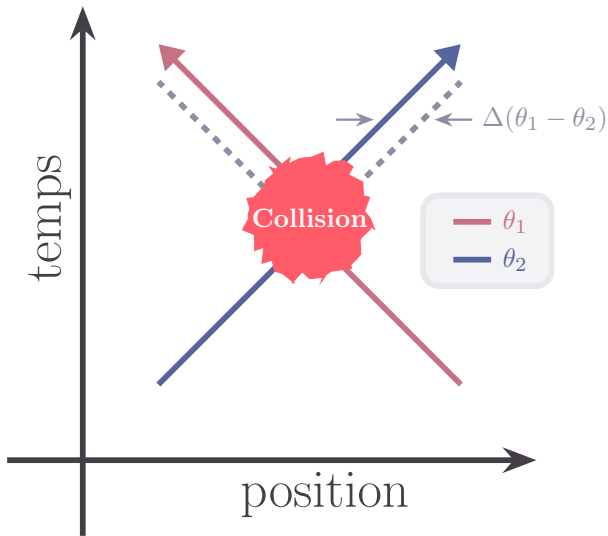
$$\int dx e(x, t) = E$$

$$\partial_t e + \partial_x j_e = 0$$

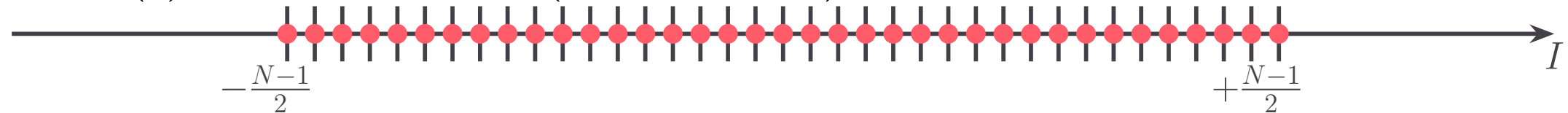




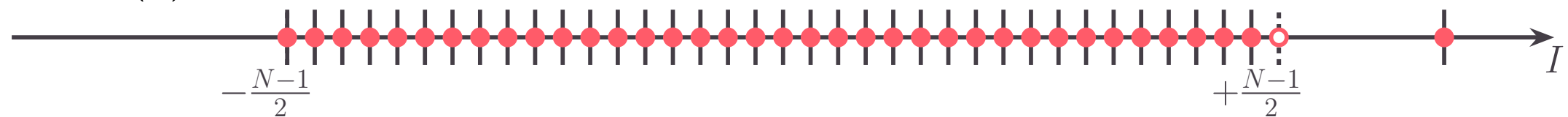




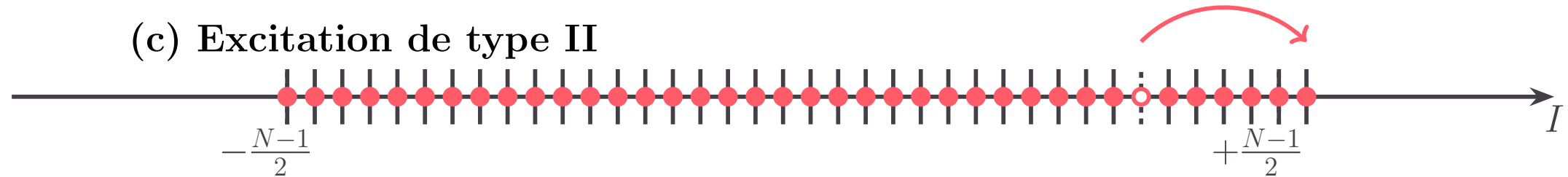
(a) État fondamental (Mer de Fermi)



(b) Excitation de type I



(c) Excitation de type II



$\bullet$: quasi-particule
 $\circ$: troue

ρ : distribution de rapidité
 ρ_h : densité de troue

$\rho_s = \rho + \rho_h$: densité d'état

